Jacob Kurry

jacobkurry1@gmail.com, (720) 305-8967, Boulder, CO GitHub link

Technical Skills

Languages: Go, Java, C/C++, Python, SQL, TypeScript (React), C#, Bash, Protobuf, Node.js
Data/ML: PyTorch, Pandas, NumPy, scikit-learn, OpenCV, RAG, Vector Search (BM25), ETL, NoSQL, Matplotlib, Redis
Container & Cloud: Kubernetes, Docker, AWS (Fargate, EC2, S3, EKS), Azure (Functions, Blob Storage, Event Grid, Function Applications, Search), Terraform, Helm, Istio
DevOps & Tools: Spring, Git, ArgoCD, FluxCD, Bamboo, Jenkins, Kubernetes Operators/CRDs, Kafka, GitLab, Kind
Testing & Observability: PyTest, Cucumber, Gherkin, Mock, JUnit, Jest, Ginkgo, Prometheus, Grafana, Loki
AI: Claude Code (subagents, skills, hooks, plugins), Model Context Protocol (MCP) servers, agentic workflows, Azure OpenAI, Cursor

Education

University of Colorado Boulder **May, 2027**
B.S., Computer Science, French *GPA*: 3.937
Relevant Coursework: Object Oriented Design, Graduate Machine Learning, Artificial Intelligence, Algorithms, Linear Algebra, Networking Systems

Internship Experience

Lunar Outpost
May 2026 – Aug 2026
Stargate Ground Software - Infrastructure Intern
Golden, CO

- Architected a **Go** network adapter routing all command and telemetry traffic between mission control and the lunar surface systems across multiple protocols; achieved **~10µs latency**, with goroutine/channel-based concurrency; added **S3** backed file-state monitoring.
- Designed and owned the full lifecycle of a custom **Link CRD** for elastic, runtime-provisioned spacecraft networking; extended the **Kubernetes** controller to actively probe live infrastructure and verify control/data-plane parity.
- Built a **Kubernetes-native operator** managing **5 CRD types** to provision spacecraft operational infrastructure (**Istio**, **Prometheus**, **Redpanda**, S3 volumes); with custom reconciliation logic, validating/mutating webhooks, and a **Kind** end-to-end test suite, and a **custom CLI** for managing and interacting with spacecraft infrastructure.
- Refactored a **TypeScript** K8s client to generate, submit, and monitor custom CRDs in real time; integrated **ArgoCD/Flux** GitOps, authored **Helm** charts; added a **Bamboo** CI pipelines for development and mission-cluster deployments.
- Created a **sidecar**, enabled via a field on the custom CRD, that watches **ConfigMaps** and applies changes to running pods in ~1s instead of waiting on the kubelet's ConfigMap sync and a Bamboo pipeline to build and push the sidecar image to **ECR**
- Built an **OpenCV**-based stereo image rectification and depth-mapping plugin for the LV2 mission, calibrating stereo camera pairs to generate depth maps and navigable path data surfaced on the mission front end.

Pure Fishing
Sep 2025 – Dec 2025
AI Development Intern – Team lead
Remote

- Architected and deployed a knowledge base system on **Azure** to extract siloed knowledge from stakeholders, implementing a three-part **serverless** solution with an ingestion/custodial platform and **RAG** application.

- Engineered **20** multi-modal data processing **pipelines** using **Azure AI** services to transform unstructured data of any type into searchable embeddings for RAG chatbot and semantic search using **BM25** + **Vector** layer.
- Implemented **Azure Functions** using **REST** and **Event grid** to manage ETL and ingestion layers.
- Built a **React** front end in **TypeScript** for the chatbot, semantic search, and file ingestion, giving stakeholders a unified interface to query and manage the knowledge base.
- Presented the final system to the executive team and won **Best Intern Project** and a trip to Charleston.

United Launch Alliance
June 2025 – Aug 2025
Ground Software Engineering Intern
Centennial, CO

- Automated engine-simulation verification with 15 **PyTest modules** on the internal CATT API; validated **2k+** discrete/analog datapoints across redline/abort/tare scenarios, reducing manual QA testing by **eight hours**.
- Developed a modular regression-testing framework (I/O abstractions, centralized logging, utilities, coding standards) integrated with **Bamboo CI**, enabling **nightly** regression testing**.**
- Improved CATT-command execution latency by **3x** through non-blocking waits and polling, maintaining backward compatibility and achieving **90% unit testing coverage** with PyTest parameterized fixtures and Mock

Personal Projects

Market ETL Pipeline
yfinance Analytics Platform

- End-to-end **ETL** pipeline to extract, transform, and load historical stock market data using **yfinance** API.
- Automated data collection, calculated key financial metrics (e.g., moving averages, P/E ratios), and stored cleaned data in structured tables using **DuckDB** for analysis.
- Applied **Python** and data processing libraries (**pandas, NumPy**) to build scalable and maintainable workflows.

Implemented API integration and data transformation workflows, orchestrating daily **CI** runs via **GitHub CI**

Wildfire Risk Prediction Model
Python, scikit-learn

- Built a machine learning model to forecast large-wildfire risk across the western U.S., using **random forest** and **gradient-boosted** classifiers to predict fire occurrences and a daily risk score.
- Engineered a data pipeline that aligns GridMet weather, LANDFIRE terrain and fuel data, MODIS satellite vegetation imagery, and MTBS burn records onto a common spatial grid and daily time step, with derived drought and rolling humidity/precipitation features.
- Designed a medallion-style data warehouse with raw and processed layers, storing versioned ingestion scripts on GitHub.